

Slope of line $2x + 3y + 4 = 0$

$$\Rightarrow m_2 = -\frac{2}{3}$$

$$\therefore \tan \theta = \left| \frac{-\frac{1}{2} + \frac{2}{3}}{1 + \frac{1}{2} \times \frac{2}{3}} \right| = \frac{1}{8}$$

$$\begin{aligned} \therefore \tan 3\theta &= \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta} \\ &= \frac{\frac{3}{8} - \frac{1}{8^3}}{1 - \frac{3}{64}} = \frac{192 - 1}{8^3} \times \frac{8^2}{61} \\ &= \frac{191}{8 \times 61} \end{aligned}$$

$$\begin{aligned} \text{Now, } \tan 3\theta &= 488 \times \frac{191}{8 \times 61} \\ &= 191 \end{aligned}$$

105. Option (a) is correct.

$$\therefore \frac{2X - 3Y}{5X + 4Y} = 4$$

$$\Rightarrow 2X - 3Y = 20X + 16Y$$

$$\Rightarrow Y = -\frac{18X}{19}$$

$$\begin{aligned} \text{Var}(X) &= npq = np(1-p) \\ &= 10 \times \frac{1}{2} \times \frac{1}{2} \end{aligned}$$

$$\Rightarrow \text{Var}(X) = \frac{5}{2}$$

$$\begin{aligned} \text{Now, } \text{Var}(Y) &= \text{Var}\left(-\frac{18}{19}X\right) \\ &= \left(-\frac{18}{19}\right)^2 \text{Var}(X) \end{aligned}$$

$$\begin{aligned} \Rightarrow \text{Var}(Y) &= \left(\frac{18}{19}\right)^2 \times \frac{5}{2} \\ &= \frac{810}{361} \end{aligned}$$

106. Option (b) is correct.

$\therefore a, b, c$ are in H.P.

$$\Rightarrow \frac{2}{b} = \frac{1}{a} + \frac{1}{c}$$

$$\Rightarrow b = \frac{2ac}{a+c}$$

$$b - a = \frac{2ac}{a+c} - a = \frac{2ac - a^2 - ac}{a+c}$$

$$= \frac{ac - a^2}{a+c} = \frac{a(c-a)}{a+c}$$

$$\Rightarrow \frac{1}{b-a} = \frac{a+c}{-a(a-c)}$$

$$\text{Now, } b - c = \frac{2ac}{a+c} - c = \frac{2ac - ac - c^2}{a+c}$$

$$= \frac{ac - c^2}{a+c} = \frac{c(a-c)}{a+c}$$

$$\Rightarrow \frac{1}{b-c} = \frac{a+c}{c(a-c)}$$

$$\therefore \frac{1}{b-a} + \frac{1}{b-c} = \frac{a+c}{-a(a-c)} + \frac{a+c}{c(a-c)}$$

$$= \frac{(a+c)}{(a-c)} \left[-\frac{1}{a} + \frac{1}{c} \right]$$

$$= \frac{a+c}{ac} = \frac{1}{a} + \frac{1}{c}$$

$$= \frac{2}{b}$$

$$= \frac{1}{b} + \frac{1}{b} = \frac{1}{b} + \frac{1}{2} \left(\frac{1}{a} + \frac{1}{c} \right)$$

So, only statement 1 and 2 are correct and statement 3 is not correct.

$\therefore$ option (b) matches the given criterion.

107. Option (c) is correct.

The rates of the edible oil in four consecutive years are 150, 200, 250, 300

$$\text{Now, average price of oil} = \frac{150 + 200 + 250 + 300}{4}$$

$$= \frac{900}{4}$$

$$= 225$$

108. Option (b) is correct.

The given word is 'TIRUPATI'.

Let E be the event of being both the T's together.

$$\therefore n(E) = 7!$$

Now, the required probability is

$$P(E) = \frac{n[E]}{n[S]} = \frac{2!7!}{8!}$$

$$= \frac{2}{8} = \frac{1}{4}$$

109. Option (c) is correct.

$$m = 77^n$$

Let us find some values of m of different values of n .

$$\text{For } n = 1, m = 77^1 = 77$$